

Hands-on Subnetting Practical

Task – 1 : Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.

Solution : **Divide 192.168.10.0/24 into 4 Equal Subnets**

Step 1: Calculate the new prefix

A /24 network has : 192.168.10.0/24

We need **4 subnets**.

Since : $2^2 = 4$

we need to borrow **2 bits** from the host portion.

Original: /24

Borrowed: 2 bits

New: /26

So the new subnet mask is:

255.255.255.192

Step 2: Calculate hosts per subnet

A /26 leaves : $32 - 26 = 6$ host bits

Therefore:

$2^6 = 64$ total addresses

$64 - 2 = 62$ usable host addresses

We subtract 2 because the first address is the **network address** and the last is the **broadcast address**.

Subnet Table

Subnet	Network Address	Usable IP Range	Broadcast Address	Usable Hosts
Subnet 1	192.168.10.0/26	192.168.10.1 – 192.168.10.62	192.168.10.63	62
Subnet 2	192.168.10.64/26	192.168.10.65 – 192.168.10.126	192.168.10.127	62
Subnet 3	192.168.10.128/26	192.168.10.129 – 192.168.10.190	192.168.10.191	62
Subnet 4	192.168.10.192/26	192.168.10.193 – 192.168.10.254	192.168.10.255	62

The pattern

The subnet increment is : $256 - 192 = 64$

Task – 2 : For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device. Hint : Make sure not to use the network or broadcast address for any device.

Solution : **Subnet Plan Diagram**

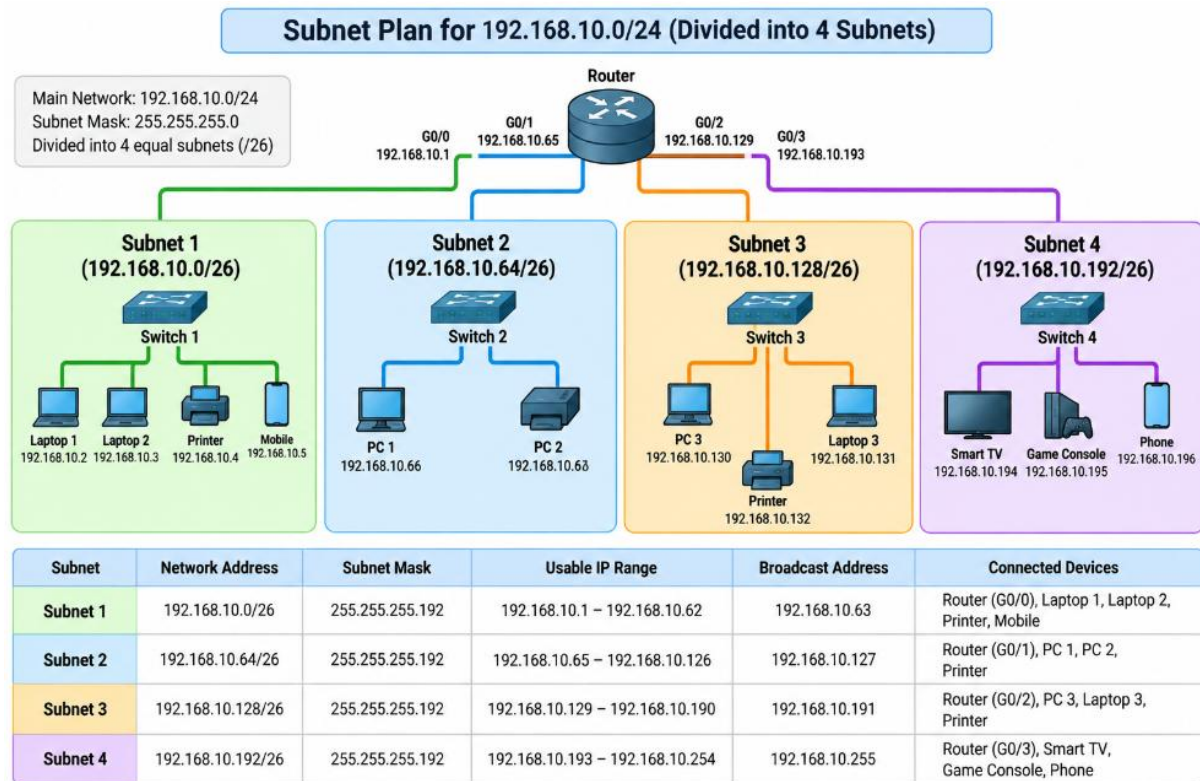
Subnet 1: 192.168.10.0/26.

- **Subnet Mask:** 255.255.255.192
- **Network Address:** 192.168.10.0
- **Usable IP Range:** 192.168.10.1 – 192.168.10.62
- **Broadcast Address:** 192.168.10.63

Device	IP Address
Router	192.168.10.1
Laptop 1	192.168.10.2
Laptop 2	192.168.10.3
Printer	192.168.10.4
Mobile Phone	192.168.10.5

Task – 3 : Draw a simple diagram (hand-drawn or digital) showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.

Solution :



Task – 4 : Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.

Solution : Subnetting Explanation

Step-by-Step

I started with the network 192.168.10.0/24. A /24 network has 256 total IP addresses, from 192.168.10.0 to 192.168.10.255.

1. I needed to create 4 equal subnets. Since $2^2 = 4$, I borrowed 2 bits from the host portion of the address.
2. The new prefix therefore becomes /26. The subnet mask for /26 is 255.255.255.192.
3. A /26 subnet contains 64 total addresses. Two addresses cannot normally be assigned to devices: the network address and the broadcast address. Therefore, each subnet has 62 usable host addresses.
4. The subnet increment is $256 - 192 = 64$. Therefore, the four network addresses are 192.168.10.0, 192.168.10.64, 192.168.10.128, and 192.168.10.192.
5. For the first subnet, 192.168.10.0/26, the usable host range is 192.168.10.1 to 192.168.10.62, and the broadcast address is 192.168.10.63.
6. I assigned 192.168.10.1 to the router, followed by 192.168.10.2 and .3 for two laptops, .4 for the printer, and .5 for the mobile device. All these addresses are valid because they fall within the usable host range.
7. The router's IP address, 192.168.10.1, can be configured as the default gateway for the other devices on this subnet.